

1. Normalize the following blog database and write the DDL scripts to create the database tables:

The screenshot shows the SQL Developer interface with two SQL files open. The first file, 'SQL File 5', contains the DDL scripts for creating the 'Authors' and 'Posts' tables. The second file, 'SQL File 3*', shows the execution output of these scripts.

```

1 CREATE TABLE Authors (
2     author_id INT AUTO_INCREMENT,
3     name VARCHAR(100) NOT NULL,
4     PRIMARY KEY(author_id)
5 );
6
7 CREATE TABLE Posts (
8     post_id INT AUTO_INCREMENT ,
9     title VARCHAR(200) NOT NULL,
10    word_count INT,
11    views INT,
12    author_id INT,
13    PRIMARY KEY (post_id),
14    FOREIGN KEY (author_id) REFERENCES Authors(author_id)
15 );
16
17

```

The output window shows the following results:

#	Time	Action	Message
1	22:18:39	CREATE TABLE Authors (author_id INT AUTO_INCREMENT, name VARCHAR(100) NOT N...	0 row(s) affected
2	22:18:39	CREATE TABLE Posts (post_id INT AUTO_INCREMENT , title VARCHAR(200) NOT NULL, ...	0 row(s) affected

The screenshot shows the SQL Developer interface with two SQL files open. The first file, 'SQL File 5', contains the INSERT statements for the 'Authors' and 'Posts' tables. The second file, 'SQL File 3*', shows the execution output of these statements.

```

1 INSERT INTO Authors (name) VALUES
2 ('Maria Charlotte'),
3 ('Juan Perez'),
4 ('Gemma Alcocer');
5
6 INSERT INTO Posts (title, word_count, views, author_id) VALUES
7 ('Best Paint Colors', 814, 14, 1),
8 ('Small Space Decorating Tips', 1146, 221, 2),
9 ('Hot Accessories', 986, 105, 1),
10 ('Mixing Textures', 765, 22, 1),
11 ('Kitchen Refresh', 1242, 307, 2),
12 ('Homemade Art Hacks', 1002, 193, 1),
13 ('Refinishing Wood Floors', 1571, 7542, 3);
14

```

The output window shows the following results:

#	Time	Action	Message	Duration / F
1	22:18:39	CREATE TABLE Authors (author_id INT AUTO_INCREMENT, name VARCHAR(100) NOT N...	0 row(s) affected	0.062 sec
2	22:18:39	CREATE TABLE Posts (post_id INT AUTO_INCREMENT , title VARCHAR(200) NOT NULL, ...	0 row(s) affected	0.063 sec
3	22:19:35	INSERT INTO Authors (name) VALUES (Maria Charlotte), (Juan Perez), (Gemma Alcocer)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
4	22:19:35	INSERT INTO Posts (title, word_count, views, author_id) VALUES (Best Paint Colors', 814, 14, 1), (...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.016 sec

2. Normalize the following airline database and write the DDL scripts to create the database tables:

gator SQL File 5 SQL File 3*

iter objects

ironhack

lab_sql

Tables

authors

posts

Views

Stored Procedures

Functions

sys

chema: lab_sql

```
1
2 CREATE TABLE Customers (
3     customer_id INT AUTO_INCREMENT PRIMARY KEY,
4     name VARCHAR(100) NOT NULL,
5     status VARCHAR(20),
6     total_mileage INT
7 );
8
9 CREATE TABLE Aircrafts (
10    aircraft_id INT AUTO_INCREMENT PRIMARY KEY,
11    name VARCHAR(50) NOT NULL,
12    total_seats INT
13 );
14
15 CREATE TABLE Flights (
16    flight_id INT AUTO_INCREMENT PRIMARY KEY,
17    flight_number VARCHAR(10) NOT NULL UNIQUE,
18    aircraft_id INT,
19    mileage INT,
20    FOREIGN KEY (aircraft_id) REFERENCES Aircrafts(aircraft_id)
21 );
22
23 -- Relación entre cliente y vuelo
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	22:31:37	CREATE TABLE Customers (customer_id INT AUTO_INCREMENT PRIMARY KEY, ...	Error Code: 1050. Table 'customers' already exists	0.000 sec
13	22:32:09	drop table customers	0 row(s) affected	0.063 sec

ct Info Session

SQL File 5 SQL File 3*

Limit to 1000 rows

```
1
2 INSERT INTO Aircrafts (name, total_seats) VALUES
3 ('Boeing 747', 400),
4 ('Airbus A330', 236),
5 ('Boeing 777', 264);
6
7
```

gator SQL File 5 SQL File 3*

iter objects

ironhack

lab_sql

Tables

aircrafts

Columns

- aircraft_id
- name
- total_seats

Indexes

Foreign Keys

Triggers

authors

flights

customers

customerflights

posts

Views

Stored Procedures

Functions

chema: lab_sql

table: aircrafts

columns:

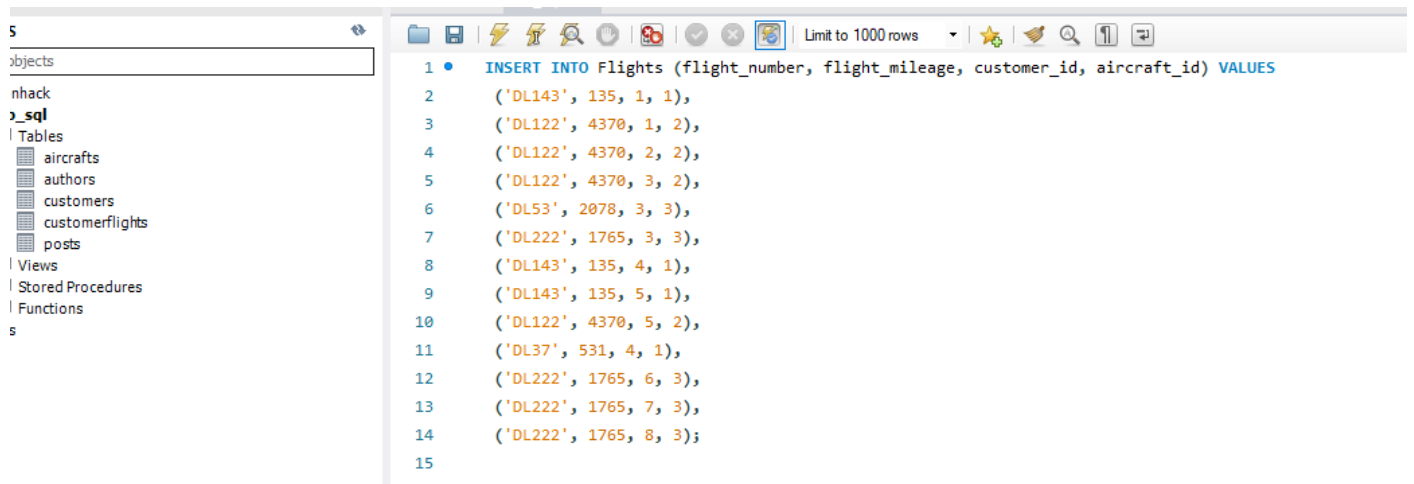
- aircraft_id int AI PK
- name varchar(50)
- total_seats int

```
1 INSERT INTO Customers (name, status, total_mileage) VALUES
2 ('Agustine Riviera', 'Silver', 115235),
3 ('Alaina Sepulvida', 'None', 6008),
4 ('Tom Jones', 'Gold', 205767),
5 ('Sam Rio', 'None', 2653),
6 ('Jessica James', 'Silver', 127656),
7 ('Ana Janco', 'Silver', 136773),
8 ('Jennifer Cortez', 'Gold', 300582),
9 ('Christian Janco', 'Silver', 14642);
10
```

Output

Action Output

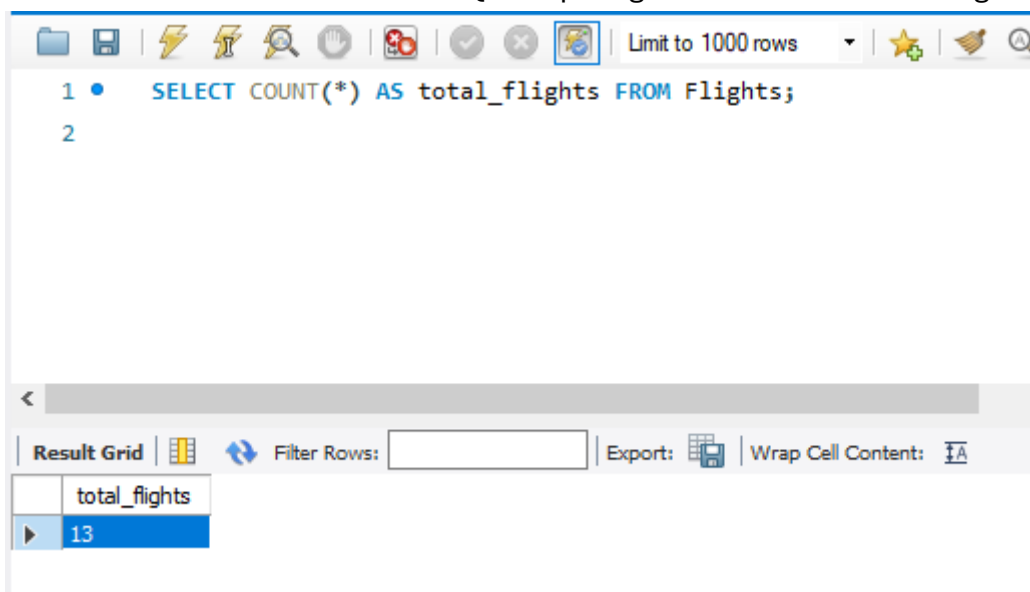
#	Time	Action	Message	Duration / Fetch
8	22:30:10	CREATE TABLE Customers (customer_id INT AUTO_INCREMENT P...	0 row(s) affected	0.078 sec
9	22:30:10	CREATE TABLE Aircrafts (aircraft_id INT AUTO_INCREMENT PRIMA...	Error Code: 1050. Table 'aircrafts' already exists	0.000 sec
10	22:31:20	DROP TABLE Aircrafts	0 row(s) affected	0.047 sec
11	22:31:20	DROP TABLE Customer	Error Code: 1051. Unknown table 'lab_sql.customer'	0.000 sec
12	22:31:37	CREATE TABLE Customers (customer_id INT AUTO_INCREMENT P...	Error Code: 1050. Table 'customers' already exists	0.000 sec
13	22:32:09	drop table customers	0 row(s) affected	0.063 sec
14	22:32:15	CREATE TABLE Customers (customer_id INT AUTO_INCREMENT P...	0 row(s) affected	0.063 sec
15	22:32:15	CREATE TABLE Aircrafts (aircraft_id INT AUTO_INCREMENT PRIMA...	0 row(s) affected	0.079 sec
16	22:32:15	CREATE TABLE Flights (flight_id INT AUTO_INCREMENT PRIMARY ...	0 row(s) affected	0.094 sec



The screenshot shows a SQL IDE with a left sidebar containing a tree view of database objects: 'objects', 'nhack', 'sql', 'Tables' (with sub-items: aircrafts, authors, customers, customerflights, posts), 'Views', 'Stored Procedures', and 'Functions'. The main editor displays an SQL script with 15 lines. The script is an INSERT statement into the 'Flights' table, with columns 'flight_number', 'flight_mileage', 'customer_id', and 'aircraft_id'. The values are provided in a list of 15 rows. The script is as follows:

```
1 • INSERT INTO Flights (flight_number, flight_mileage, customer_id, aircraft_id) VALUES
2 ('DL143', 135, 1, 1),
3 ('DL122', 4370, 1, 2),
4 ('DL122', 4370, 2, 2),
5 ('DL122', 4370, 3, 2),
6 ('DL53', 2078, 3, 3),
7 ('DL222', 1765, 3, 3),
8 ('DL143', 135, 4, 1),
9 ('DL143', 135, 5, 1),
10 ('DL122', 4370, 5, 2),
11 ('DL37', 531, 4, 1),
12 ('DL222', 1765, 6, 3),
13 ('DL222', 1765, 7, 3),
14 ('DL222', 1765, 8, 3);
15
```

3. In the Airline database write the SQL script to get the total number of flights in the database.



The screenshot shows a SQL IDE with a toolbar at the top. The main editor displays an SQL script with 2 lines. The script is a SELECT statement that counts the total number of flights in the 'Flights' table. The script is as follows:

```
1 • SELECT COUNT(*) AS total_flights FROM Flights;
2
```

Below the editor, there is a 'Result Grid' section. It shows a single row with the column name 'total_flights' and the value '13'.

total_flights
13

4. In the Airline database write the SQL script to get the average flight distance.

The screenshot shows a SQL query editor with a toolbar at the top. The query is: `SELECT AVG(flight_mileage) AS avg_flight_distance FROM Flights;`. Below the query editor, the 'Result Grid' is displayed with the following data:

avg_flight_distance
2119.5385

5. In the Airline database write the SQL script to get the average number of seats.

The screenshot shows a SQL query editor with a toolbar at the top. The query is: `SELECT AVG(total_seats) AS avg_seats FROM Aircrafts;`. Below the query editor, the 'Result Grid' is displayed with the following data:

avg_seats
300.0000

6. In the Airline database write the SQL script to get the average number of miles flown by customers grouped by status.

SQL File 5 lab_sql lab_sql.customerflights

Limit to 1000 rows

```

1 • SELECT status, AVG(total_mileage) AS avg_total_mileage
2   FROM Customers
3  GROUP BY status;
4

```

Result Grid

	status	avg_total_mileage
▶	Silver	98576.5000
	None	4330.5000
	Gold	253174.5000

7. In the Airline database write the SQL script to get the maximum number of miles flown by customers grouped by status.

Limit to 1000 rows

```

1 • SELECT status, MAX(total_mileage) AS max_total_mileage
2   FROM Customers
3  GROUP BY status;
4

```

Result Grid

	status	max_total_mileage
▶	Silver	136773
	None	6008
	Gold	300582

8. In the Airline database write the SQL script to get the total number of aircraft with a name containing Boeing.

Limit to 1000 rows

```

1 • SELECT COUNT(*) AS total_boeing_aircrafts
2   FROM Aircrafts
3  WHERE name LIKE '%Boeing%';

```

Result Grid

total_boeing_aircrafts
2

9. In the Airline database write the SQL script to find all flights with a distance between 300 and 2000 miles.

Limit to 1000 rows

```

1 • SELECT * FROM Flights
2   WHERE flight_mileage BETWEEN 300 AND 2000;
3

```

Result Grid

flight_id	flight_number	flight_mileage	customer_id	aircraft_id
6	DL222	1765	3	3
10	DL37	531	4	1
11	DL222	1765	6	3
12	DL222	1765	7	3
13	DL222	1765	8	3
NULL	NULL	NULL	NULL	NULL

10. In the Airline database write the SQL script to find the average flight distance booked grouped by customer status (this should require a join).

1	•	SELECT Customers.status, AVG(Flights.flight_mileage) AS avg_milea
2		FROM Flights
3		JOIN Customers ON Flights.customer_id = Customers.customer_id
4		GROUP BY Customers.status;
5		

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	status	avg_mileage		
▶	Silver	2090.0000		
	None	1678.6667		
	Gold	2494.5000		

11. In the Airline database write the SQL script to find the most often booked aircraft by gold status members (this should require a join).

2	FROM Flights
3	JOIN Customers ON Flights.customer_id = Customers.customer_id
4	JOIN Aircrafts ON Flights.aircraft_id = Aircrafts.aircraft_id
5	WHERE Customers.status = 'Gold'
6	GROUP BY Aircrafts.name

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	name	bookings			
▶	Boeing 777	3			